Problem Statement: Smart Personal Assistant Using Multi-Agent System

Build an advanced Multi-Agent AI Assistant that can intelligently process different categories of user requests by delegating tasks to specialized agents using external APIs.

The system should accept a user query, identify the intent using a Router Agent, and forward the request to the relevant specialized agents. Each agent must independently fetch real-time data from APIs and generate structured outputs. A Combiner Agent should then merge all responses into a single final response.

Additionally, the system must generate a timestamped `.txt` report file containing:

- User Query
- Agent Responses
- API Outputs
- Final Combined Response
- Execution Date & Time

The generated report should be automatically stored for future reference or auditing purposes.

Agents to Implement

1. Weather Agent

Responsible for providing current weather information using the Open-Meteo API.

API

- https://api.open-meteo.com/v1/forecast?latitude=13.08&longitude=80.27&current_weather=true

Example Queries

- “What’s the weather today?”
- “Will it rain today?”

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2. Crypto Price Agent

Fetches live cryptocurrency prices such as Bitcoin, Ethereum, and Dogecoin.

API

- https://api.coingecko.com/api/v3/simple/price?ids=bitcoin,ethereum,dogecoin&vs_currencies=usd,inr

Example Queries

- “What is Bitcoin price?”
- “Show Ethereum and Dogecoin prices.”

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3. Currency Exchange Agent

Handles currency conversion and exchange-rate retrieval.

API

- https://open.er-api.com/v6/latest/USD

Example Queries

- “Convert USD to INR”
- “Show latest exchange rates”

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4. Joke Agent

Generates random jokes for entertainment.

API

- https://v2.jokeapi.dev/joke/Any

Example Queries

- “Tell me a joke”
- “Give me something funny”

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5. Quote Agent

Provides motivational and inspirational quotes.

API

- https://api.quotable.io/random

Example Queries

- “Give me a motivational quote”
- “Inspire me”

Workflow Requirements

Router Agent

- Detect user intent
- Decide which agents should execute
- Support multiple agent activations for a single query

Specialized Agents

Each agent must:

- Call external APIs
- Process API responses
- Return structured outputs
- Handle failures using retries

Combiner Agent

The Combiner Agent should:

- Collect outputs from all executed agents
- Merge results into a single coherent response
- Remove redundant information
- Format outputs cleanly for end users

Report Generator Module

After generating the final response:

- Create a `.txt` file automatically
- Append current date and time to filename
- Store:
    - User query
    - Selected agents
    - Raw API outputs
    - Final summarized response
    - Execution timestamp

Example Filename

- report_2026_05_18_14_30.txt

Additional System Features

- Shared State Management
- Retry Mechanism
- JSON Output Storage
- Workflow Visualization
- Modular Agent Architecture
- Extensible Tool Integration

Expected Outcome

The final system should function as a real-world intelligent AI assistant capable of:

- Real-time API integrations
- Intelligent task delegation
- Multi-agent collaboration

- Structured workflow orchestration
- Dynamic response generation
- Automated report generation
- Scalable AI system design